

Sponsor Engagement Stakeholder Analysis Effective Interviewing

DAEN 459/460 Capstone Senior Design

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ENGINEERING
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The Goal is: Shared Confidence

Not Just Frequent Contact

- Set a sponsor cadence that produces decisions
- Identify whose needs can reshape the solution
- Run conversation/interviews that reveal **workflow**, **constraints**, and **evidence**
- Convert conversations into testable requirements



A useful sponsor meeting ends with:

- 1) Decisions
- 2) Owners
- 3) Evidence needed
- 4) Next checkpoint

Example

Riverton Public Works (Sponsor)

- **Case**

Reduce missed pickups without creating surveillance

- Use service-request, route, and weather data to help supervisors anticipate missed pickups.

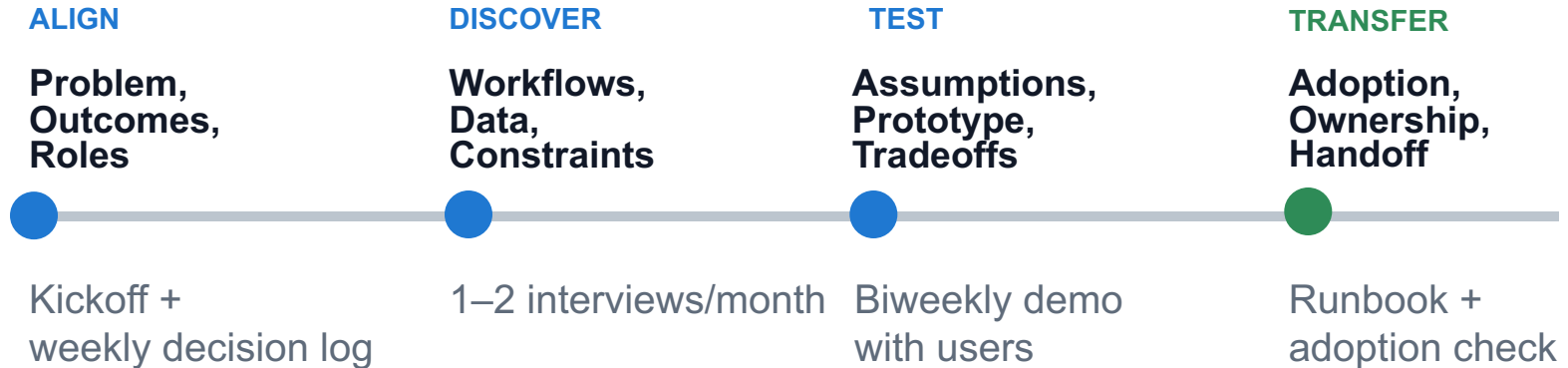
- **First request:**

"Build us a dashboard that predicts tomorrow's problem routes."

What is Still Unknown?

- Who acts on a prediction?
- What counts as a "miss"?
- Which errors are unacceptable?
- What data can legally be used?
- How will success be judged?

Use Sponsor Time to Reduce Uncertainty in Stages



Rule

Every interaction should close an uncertainty, confirm a decision, or expose a new risk.

Agree on the "Working Contract"

Before the Work Accelerates

DECISION RIGHTS	What can the team decide?
AVAILABILITY	Who joins which meeting? Who may be the backup?
FEEDBACK WINDOW	When is silence treated as approval?
ESCALATION PATH	What happens when scope, data, or access stalls?

Map Influence x Impact

(Then Plan Distinct Engagement)

For each stakeholder:

- Desired outcome
- Fears / constraints
- Evidence they trust
- Decision power
- Best channel

A Stakeholder Register Turns a Map into an Engagement Plan

Stakeholder	What They Need	Project Risk	Evidence to Seek	Engagement
Public Works Director	Fewer complaints; defensible spend	Dashboard optimizes the wrong outcome	Definition of success; decision thresholds	Bi-monthly decision review
Route Supervisors	Fewer complaints; defensible spend	Alerts arrive too late or lack context	Daily workflow; override reasons	Co-design + prototype tests
Drivers	Fair, feasible assignments	Model penalizes crews for bad inputs	Exception patterns; route realities	Short interviews; ride-along
Residents	Reliable service; privacy	Sensitive data used without consent	Complaint journey; acceptable data use	Survey + representative interviews
IT /Privacy	Secure, supportable solution	Unauthorized data or unsupported stack	Access rules; retention; architecture	Gate reviews

Interview for Decisions and Behavior

(Not Feature Votes)

Weak

- *"Would a predictive dashboard help you?"*

Invites politeness and speculation

Better

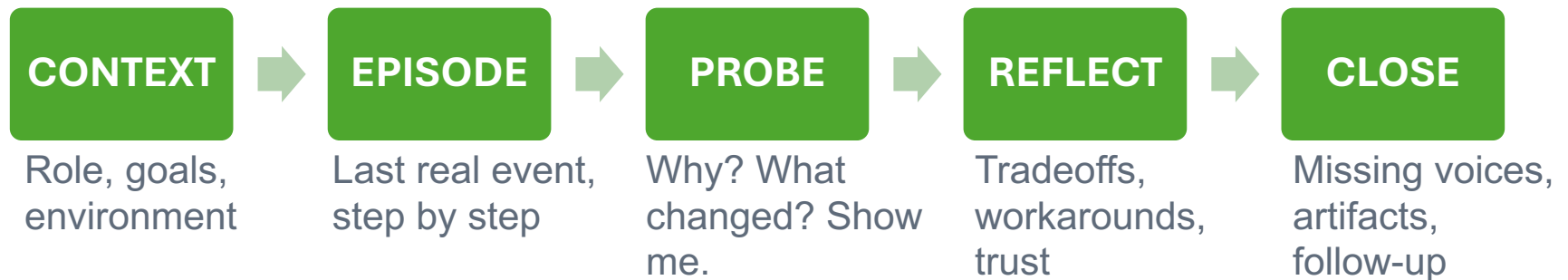
- *"Walk me through the last missed pickup you handled."*

Reveals actual workflow and evidence

Target:

**Concrete Episodes → Observed Constraints →
Decision Criteria → Testable Requirements**

Strong Interviews Follow the Story Before Testing Solutions



Listen for verbs, exceptions, handoffs, artifacts, and consequences.

Use a question ladder to move from **story** to **requirement**

1. OPEN

Tell me about the last missed pickup.

2. SEQUENCE

What happened first? Then what?

3. PROBE

What information did you use? What was missing?

4. EXCEPTION

When does this process break down?

5. CRITERION

What would make an alert trustworthy enough to act on?

Avoid: Leading Questions • Stacked Questions • Jargon • Defending *the* Solution

Example:

One answer changes the product concept

Interview Excerpt

Supervisor: *"By the time I see the complaint report, the truck is already across town. I usually call the driver because the route system does not show blocked alleys."*

SIGNAL	Complaint report is delayed
WORKAROUND	Supervisor calls the driver
HIDDEN CONTEXT	Blocked alleys are absent from route data
REQUIREMENT	Surface operational exceptions before dispatch
TEST	Can a supervisor act in under 2 minutes?

The pivot:

From predicting problem routes

To supporting exception-aware dispatch

Synthesize Immediately:

Evidence, Interpretation, Decision

EVIDENCE

**Exact observation
or short quote
+ source and date**

Do not jump

Quote → feature

INTERPRETATION

**Pattern, tension,
or unmet need
+ confidence level**

Build the chain

Evidence → insight → decision → test

ACTION

**Requirement,
experiment, or
question
+ owner and deadline**

Practice:

Interview the Route Supervisor

INTERVIEWER

**Use the five-step arc.
Ask for one real
episode.**

SUPERVISOR

**You distrust alerts
without route
context. Reveal this
only if probed.**

OBSERVER

**Track leading
questions, follow-
ups, and usable
evidence.**

Debrief:

**What changed in your understanding?
What requirement can you now test?**

Next Sponsor Meeting:

Leave with Four Artifacts

1. Working Contract

Decision rights, cadence, feedback, escalation

2. Stakeholder Register

Influence, impact, needs, evidence, channel

3. Interview Guide

Purpose, arc, probes, closing questions

4. Decision Log

Evidence, decision, owner, due date!

Evidence → Insight → Decision → Test